Packet Tracer – EIGRP Routing in a Multi-Router Topology

Objectives

Part 1: Build a Multi-Router Network Topology
Part 2: Configure IP Addressing
Part 3: Configure EIGRP Routing
Part 4: Verify EIGRP Neighbor Relationships
Part 5: Verify Routing Tables
Part 6: Test End-to-End Connectivity

Background / Scenario

You are the network administrator for a company with multiple branch offices connected through routers. Instead of manually creating static routes on every router, you decide to implement **Enhanced Interior Gateway Routing Protocol (EIGRP)**.

EIGRP is a Cisco proprietary dynamic routing protocol that automatically shares network information between routers. When properly configured, routers learn remote networks automatically and update routing tables without requiring manual route entries.

In this lab, you will:

- Configure a three-router topology
- Configure EIGRP Autonomous System (AS)
- Verify neighbor relationships
- Examine dynamically learned routes
- Test connectivity between remote networks

This lab demonstrates how dynamic routing simplifies network management compared to static routing.

Instructions

Part 1: Build the Network Topology

Step 1: Add the Devices

a. Open **Cisco Packet Tracer**.

b. Add:

- 3 Routers (2911)
- 2 PCs

c. Rename devices:

- Router0 → **R1**
- Router1 → **R2**
- Router2 → **R3**
- PC0 → **PC-A**
- PC1 → **PC-B**

Step 2: Connect the Devices

Use the following connections:

Device	Port	Device	Port
PC-A	FastEthernet0	R1	GigabitEthernet0/0
R1	GigabitEthernet0/1	R2	GigabitEthernet0/0
R2	GigabitEthernet0/1	R3	GigabitEthernet0/0
PC-B	FastEthernet0	R3	GigabitEthernet0/1

Use **Copper Straight-Through** cables for all connections.

Part 2: Configure IP Addressing

Step 1: Configure PC-A

Desktop → IP Configuration

Enter:

- IP Address: 192.168.10.10

- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

Step 2: Configure PC-B

Desktop → IP Configuration

Enter:

- IP Address: 192.168.30.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.30.1

Step 3: Configure R1

```
enable
configure terminal

interface g0/0
 ip address 192.168.10.1 255.255.255.0
 no shutdown

interface g0/1
 ip address 10.1.12.1 255.255.255.252
 no shutdown

end
```

Step 4: Configure R2

```
enable
configure terminal

interface g0/0
 ip address 10.1.12.2 255.255.255.252
 no shutdown
```

```
interface g0/1
 ip address 10.1.23.1 255.255.255.252
 no shutdown

end
```

Step 5: Configure R3

```
enable
configure terminal

interface g0/0
 ip address 10.1.23.2 255.255.255.252
 no shutdown

interface g0/1
 ip address 192.168.30.1 255.255.255.0
 no shutdown

end
```

Part 3: Configure EIGRP

Why EIGRP?

Without EIGRP:

R1 knows:
192.168.10.0
10.1.12.0

R1 does NOT know:
10.1.23.0
192.168.30.0

EIGRP automatically advertises networks and learns remote routes.

Step 1: Configure EIGRP on R1

enable
configure terminal

router eigrp 100

network 192.168.10.0 0.0.0.255
network 10.1.12.0 0.0.0.3

no auto-summary

end

Step 2: Configure EIGRP on R2

enable
configure terminal

router eigrp 100

network 10.1.12.0 0.0.0.3
network 10.1.23.0 0.0.0.3

no auto-summary

end

Step 3: Configure EIGRP on R3

enable
configure terminal

router eigrp 100

network 10.1.23.0 0.0.0.3
network 192.168.30.0 0.0.0.255

no auto-summary

end

Important Note

All routers must use the same AS number:

router eigrp 100

If AS numbers differ:

R1 = 100
R2 = 200

No EIGRP neighbors will form.

Part 4: Verify EIGRP Neighbors

Check R1 Neighbor Table

show ip eigrp neighbors

Expected:

Address Interface
10.1.12.2 Gi0/1

Check R2 Neighbor Table

show ip eigrp neighbors

Expected:

Address Interface

10.1.12.1 Gi0/0
10.1.23.2 Gi0/1

Check R3 Neighbor Table

show ip eigrp neighbors

Expected:

Address Interface

10.1.23.1 Gi0/0

Part 5: Verify Routing Tables

Check R1 Routes

show ip route

Expected:

D 10.1.23.0/30
D 192.168.30.0/24

Notice the **D**.

D = EIGRP Learned Route

Check R2 Routes

show ip route

Expected:

D 192.168.10.0/24
D 192.168.30.0/24

Check R3 Routes

show ip route

Expected:

D 192.168.10.0/24
D 10.1.12.0/30

View EIGRP Topology Table

show ip eigrp topology

Expected:

P 192.168.10.0/24
P 192.168.30.0/24

Where:

P = Passive State

Passive means:

Good
Stable
No route recalculation occurring

Part 6: Test Connectivity

Step 1: Ping Between Routers

From R1:

ping 10.1.23.2

Expected:

!!!!!

Success rate = 100%

Step 2: Ping from PC-A to PC-B

On PC-A:

ping 192.168.30.10

Expected:

Reply from 192.168.30.10

4 successful replies.

Step 3: Ping from PC-B to PC-A

On PC-B:

ping 192.168.10.10

Expected:

Reply from 192.168.10.10

4 successful replies.

Step 4: Run Traceroute

On PC-A:

tracert 192.168.30.10

Expected:

192.168.10.1
10.1.12.2
10.1.23.2
192.168.30.10

Traffic should pass:

R1 → R2 → R3

Troubleshooting Tips

If routes are missing:

✓ Verify interfaces are up

show ip interface brief

✓ Verify EIGRP neighbors exist

show ip eigrp neighbors

✓ Verify AS numbers match

show running-config

✓ Verify network statements

show running-config | section eigrp

✓ Verify routing table

show ip route

✓ Confirm all interfaces are in the correct subnet

What You Learned

- How dynamic routing differs from static routing
- How EIGRP forms neighbor relationships
- How EIGRP advertises networks automatically
- How routers learn remote networks dynamically
- How to verify EIGRP routes and topology tables
- How EIGRP simplifies large network administration
- How end-to-end connectivity is established through dynamic routing

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Challenge Extensions (Optional)

Challenge 1: Add a Fourth Router

Create:

```
PC-A
 |
R1
 |
R2
```

|
R3
|
R4
|
PC-C

Configure EIGRP and verify all routers learn routes automatically.

Challenge 2: Break a Neighbor Relationship

On R2:

router eigrp 200

Questions:

- What happens to neighbors?
- What routes disappear?
- Why?

Restore:

router eigrp 100

Challenge 3: Disable Auto-Summary

Remove:

no auto-summary

Observe route behavior.

Research:

- Why was auto-summary problematic?
- Why is it disabled in modern networks?

Challenge 4: Observe EIGRP Packets

Switch Packet Tracer to **Simulation Mode**.

Generate traffic and observe:

- EIGRP Hello packets
- Neighbor formation
- Route advertisements
- ICMP traffic

Challenge 5: Compare Static Routing vs EIGRP

Delete EIGRP configuration:

no router eigrp 100

Replace with static routes.

Questions:

- How many routes must be manually configured?
- Which method scales better?
- Which is easier to maintain?